PowerSight

Automate Meter Reading & Empower Sustainable Choices

Capstone Project 2024

Computer Science and Engineering Department
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CPG 171

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PROJECT OVERVIEW

PowerSight **transforms old meters into smart meters** using **image processing**. This captured data is then analyzed by **machine learning** to unlock valuable insights into your energy consumption. Through a user-friendly web dashboard, you gain access to **real-time monitoring**, historical trends, and **predictive forecasts,** empowering you to make informed decisions towards a more **sustainable and cost-effective energy future**.

PROBLEM STATEMENT

Despite advancements in energy monitoring technology, many regions still rely on manual meters, which lack real-time insights and predictive analysis capabilities. Converting these manual meters into smart meters presents a cost-effective solution to enhance energy monitoring and promote sustainable energy management. However, traditional methods of meter conversion are often expensive and labor-intensive, posing challenges for widespread adoption.

OBJECTIVE

Automate Meter Data Collection:

Develop an image processing solution to convert existing manual meters into smart meters, enabling automated data extraction and analysis.

Implement Predictive Power Consumption Analysis:

Integrate machine learning algorithms to provide predictive analysis and real-time insights into electricity consumption patterns, facilitating proactive energy management.

Create an Interactive User Dashboard:

Design an interactive internet dashboard to present consumption trends, real-time projections, and relevant comparisons, empowering users to make informed decisions for sustainable energy usage.

Promote Sustainable Energy Management:

Evaluate the cost-effectiveness of the proposed solution, ensuring accessibility and affordability for widespread adoption and impact.

NEED ANALYSIS

- Traditional meter reading methods are often **manual, time-consuming, and prone to errors**.

- They lack the ability to provide **detailed consumption insights** needed for informed decision-making.

- This leads to **inefficient energy management** and **missed opportunities for cost savings and sustainability improvements**.

LITERATURE SURVEY

There are many approaches for measuring the vital parameters for Automated power consumption Analysis.

- According to Smith et al. (2019), manual meter reading is error-prone and resource-intensive, leading to a rising demand for automated solutions.
- Zhang et al. (2021) showcased the efficacy of machine learning models, like support vector machines and recurrent neural networks, in predicting electricity consumption with adaptability.
- Patel et al. (2019) emphasized the effectiveness of interactive dashboards in empowering users to monitor and optimize energy consumption, aligning with the goals of Automated Power Consumption Analysis.

ASSUMPTIONS AND CONSTRAINTS

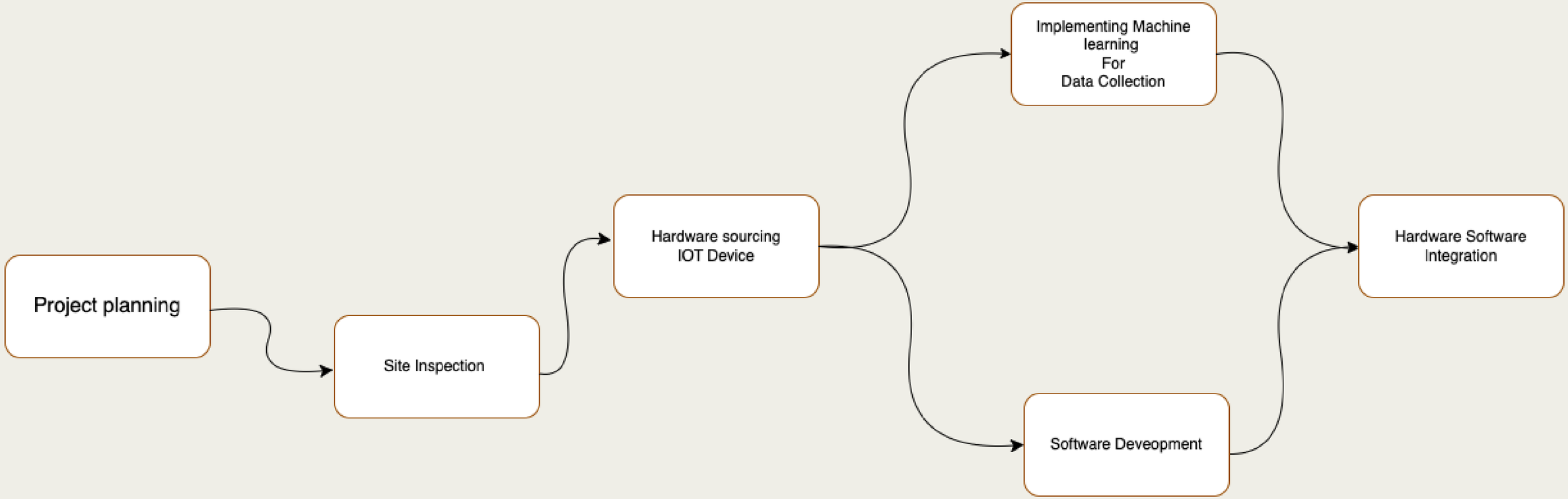
Assumptions

- Reliable Image Capture
- Consistent Meter Readings
- Accurate Algorithms
- Stable Internet
- Data Privacy

Constraints

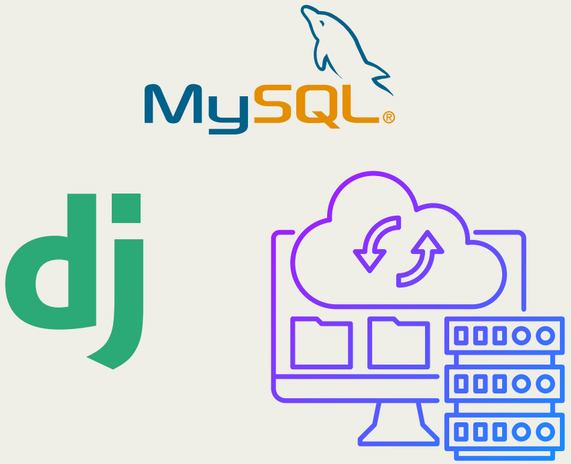
- Hardware Limitations
- Meter Compatibility
- Environmental Factors
- Regulatory Compliance
- Cost Constraints
- User Interface Design

PROJECT EXECUTION PLAN



PROJECT REQUIREMENTS

- **SOFTWARE**
  - Open CV
  - Django
  - Html/Css/Javascript
  - Python
  - Mysql
  - Cloud Services
  - User Authentication



- **HARDWARE**
  - IP Wifi Camera
  - Internet Supported Device

PROJECT OUTCOMES

Smart Meter Transformation

- Transform traditional meters into smart tools with the latest image processing technology.
- Enhance functionality, accessibility, and data accuracy.

Effortless Monitoring

- Streamline meter data collection with advanced image processing, ensuring accuracy and reducing manual effort.
- Achieve a high success rate under diverse conditions.

User-Centric Dashboard

- Access a user-friendly web dashboard for real-time and historical power consumption data.
- Visualizing trends effortlessly and gaining insights.

Empowering Sustainability

- Empowers users to optimize energy consumption and reduce costs.
- Promotes significant reduction in overall energy consumption through data-driven comparisons.

Intelligent Power Insights

- Machine learning model for precise real-time and future power consumption predictions.
- Witness the model's ongoing progress as it adapts over time to meet changing energy needs.

WORK PLAN

Phase 1: Planning & Assessment

Define project scope (meter types, data format, features). Research technology options (communication, data collection, system architecture). Estimate costs (hardware, software).

Phase 2: Design & Development

Select compatible meters/communication modules; develop software for meter interface, data collection, security, and system integration. Rigorously test functionality, accuracy, and security.

Phase 3: Deployment & Hardware Integration

Install new cameras for data collection. Integrate the system with our software.

Phase 4: Monitoring & Maintenance

Regularly monitor system performance. Analyze data to gain insights and identify optimization opportunities. Perform maintenance and apply security updates.

CONTRIBUTIONS OF INDIVIDUAL TEAM MEMBERS

ACTIVITY	NIKHIL	CHINMAYEE	TEJASWI	VASHUKI	PURU
PROJECT SELECTION PHASE	✓	✓	✓	✓	✓
PROJECT PLANNING	✓	✓	✓	✓	✓
INITIAL DOCUMENTATION	✓	✓	✓	✓	✓
ANALYSIS PHASE	✓		✓	✓	✓
FRONTEND DEVELOPMENT		✓	✓		✓
BACKEND DEVELOPMENT	✓				
MACHINE LEARNING MODEL	✓		✓		✓
DESIGNING AND DEPLOYMENT				✓	✓
HARDWARE DEPLOYMENT		✓		✓	
HARDWARE TESTING	✓			✓	
SOFTWARE TESTING	✓	✓	✓	✓	✓
FINAL REPORT	✓	✓	✓	✓	✓

Thank you!